

Random Matrices

$$\begin{pmatrix} * & \dots & * \\ \vdots & & \vdots \\ * & \dots & * \end{pmatrix} \quad \begin{array}{l} \text{These entries are random} \\ X_1 \dots X_n \text{ vectors} \in \mathbb{R}^d \\ X_i = \underbrace{(X_{i1} \dots X_{id})}_{\text{Random variables}} \end{array}$$

$$\frac{1}{n} \sum X_{i,i}^2 \longrightarrow \text{Variance}$$

$$\frac{1}{n} \sum X_{i1} X_{i2} \longrightarrow \text{Covariance}$$

$$X = \begin{pmatrix} X_{11} & \dots & X_{1d} \\ \vdots & & \vdots \\ X_{n1} & \dots & X_{nd} \end{pmatrix} \quad \text{and} \quad \frac{XX^T}{n} \quad \text{is the sample covariance matrix}$$

Eigenvalues

$$Av = \lambda v \text{ for } v \neq 0.$$

$$\text{tr}(A) = \sum A_{ii} = \sum_{i=1}^n \lambda_i$$

{ Set of eigenvalues of a random matrices }
= Empirical Spectral distribution.

$$\frac{1}{n} \sum \delta_{\lambda_i}$$

$\begin{pmatrix} A_{11} & & A_{1j} \\ & \ddots & \\ A & & A_{mn} \end{pmatrix}$ entries are prob $1/2$
for 1 or -1 .
 $A_{ji} = A_{ij}$

$$\text{tr}(A) = \sum A_{ii} = \sum \lambda_i$$

$$\text{tr}(A^2) = \sum A_{ii}^2 = \sum \lambda_i^2$$

$$\text{tr}(A^k) = \sum A_{ii}^k = \sum \lambda_i^k$$